



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science & Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: TY

Course: Elective: Advanced
Python Programming

Course Code: PEC-CS303

Date: 22-Aug-25

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- i. *Mobile phones and programmable calculators are strictly prohibited.*

Q.N.	Attempt all questions	Marks	CO	BT
1	Define class and object. Write a program for object initialization.	5	1	BT2
2	With a suitable program explain the concept of inheritance with its types.	5	1	BT2
3	With a suitable program explain the concept of encapsulation.	5	1	BT2
4	Explain NumPy library with two suitable programs.	5	4	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply advanced object-oriented programming concepts, including meta programming and design patterns.
CO 2	Implement concurrent and parallel programs using threads, processes, and asynchronous techniques.
CO 3	Develop network applications using sockets and advanced networking libraries.
CO 4	Utilize advanced Python libraries for data manipulation, scientific computing, and web development.
CO 5	Write efficient and optimized Python code for performance-critical applications.
CO 6	Design and implement robust and maintainable Python applications.



SGGS Institute of Engineering and Technology, Vishnupuri,
Nanded

Academic Year: 2025-2026

In-Semester Evaluation-I

Class: Third Year (All Branch)

Date: 25/08/25

Time: 15:00-15:45 Hrs

Subject: Introduction to Cryptography

Code: MDM-IS-CS301

Max Marks: 20

Q.N.	Questions	Marks	CO	BT																																																																																					
1.	Define cryptography and give its significance in today's world? How do symmetric and asymmetric cryptography differ?	[6]	CO2	BT1																																																																																					
2.	a. Explain the fundamental concept behind a Feistel network, including how it divides plaintext and processes it through multiple rounds. OR b. Explain 1. Differential Cryptanalysis 2. Meet in the middle attack	[8] [8]	CO2 CO2	BT1 BT2																																																																																					
3.	1) Identify valid or invalid congruences a) $7 \equiv -11 \pmod{6}$ b) $97 \equiv 19 \pmod{12}$ 2) Find the multiplicative inverse of 23 in Z_{100} 3) Using below given S-box 1 of DES , find the output for the input 110011. Show the row, column selection, and final binary result. <table><tr><td></td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td><td>15</td></tr><tr><td>0</td><td>14</td><td>04</td><td>13</td><td>01</td><td>02</td><td>15</td><td>11</td><td>08</td><td>03</td><td>10</td><td>06</td><td>12</td><td>05</td><td>09</td><td>00</td><td>07</td></tr><tr><td>1</td><td>00</td><td>15</td><td>07</td><td>04</td><td>14</td><td>02</td><td>13</td><td>10</td><td>03</td><td>06</td><td>12</td><td>11</td><td>09</td><td>05</td><td>03</td><td>08</td></tr><tr><td>2</td><td>04</td><td>01</td><td>14</td><td>08</td><td>13</td><td>06</td><td>02</td><td>11</td><td>15</td><td>12</td><td>09</td><td>07</td><td>03</td><td>10</td><td>05</td><td>00</td></tr><tr><td>3</td><td>15</td><td>12</td><td>08</td><td>02</td><td>04</td><td>09</td><td>01</td><td>07</td><td>05</td><td>11</td><td>03</td><td>14</td><td>10</td><td>00</td><td>06</td><td>13</td></tr></table>		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07	1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08	2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00	3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13	[6]	CO1 CO1 CO1	BT5 BT5 BT3
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